

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		award (1) for either of following - the groups bonded to each carbon atom of the C=C must be the same - it must be a symmetrical alkene			1	1		
		(ii)		orange / red / yellow	1			1		1
		(iii)	I	award (1) for any of following - it does not contain a $\begin{array}{c} \text{H} \\ \diagup \\ \text{C} \\ \diagdown \\ \text{O} \end{array}$ group - it is not an aldehyde			1	1		
			II	it is not an aldehyde, therefore cannot be $\text{CH}_3(\text{CH}_2)_3\text{CHO}$ or $\text{CH}_3(\text{CH}_2)_2\text{CHO}$ (1) accept it must be $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH}_3(\text{CH}_2)_3\text{C} \\ \diagdown \\ \text{O} \end{array}$ or $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH}_3\text{CH}_2\text{C} \\ \diagdown \\ \text{O} \end{array}$ melting temperature cannot be higher than the literature value therefore it cannot be $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH}_3(\text{CH}_2)_3\text{C} \\ \diagdown \\ \text{O} \end{array}$ (1) compound U must be $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH}_3\text{CH}_2\text{C} \\ \diagdown \\ \text{O} \end{array}$ (1) ecf possible	1 1					
						1		3		